

RESEARCH ARTICLE

An Antifouling Coating Based on Tannic Acid: An Eco-Friendly Strategy for Antioxidant Removal of Reactive Oxygen Species

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ABSTRACT

As a kind of natural polyphenolic antioxidants, tannic acid (TA) has a wide range of sources and biosafety, excellent antibacterial properties and its special chemical structure makes it easy to bind to the surface, which may provide inspiration for the development of antifouling materials. In this paper, a TA-based antifouling coating was designed and prepared. A stable intermediate transition layer was formed by co-deposited 3,4-dihydroxy phenyl-L-alanine (DOPA) and polyethyleneimine (PEI) on the surface of polyvinyl chloride (PVC), and then TA molecules were grafted on the surface to prepare a TA functionalized (PVC/(PDA/PEI)-TA) coating. The effect of the coating on the adhesion of marine bacteria, benthic diatoms and mussels was investigated. The results indicated that the coating had good antifouling performance. Marine bacteria and benthic diatoms had a low adhesion rate on the coating surface and were prone to the detachment under water flow erosion. The TA on the coating surface affected the adhesion behavior of mussels, which only secreted a small amount of adhesive plaques on the surface, and the adhesion force between the plaques and the substrate became low. This study may provide new solutions to marine biofouling.

1 | Introduction

Biofouling is the accumulation of aquatic organisms (e.g., microorganisms, plants, and animals) on surfaces and structures immersed in or exposed to the aquatic environment [1]. This increasingly serious problem has brought adverse impacts on maritime and aquatic industries, resulting in serious economic losses [2–5]. In recent years, coatings based on cuprous oxide biocides are widely used to prevent and control marine biofouling. However, the non-target toxicity of these biocides causes serious ecological hazards to marine systems [1, 6, 7]. Hence, eco-friendly antifouling technologies have emerged as the times require, including surface hydration (e.g., polyethylene glycol, zwitterionic polymers) [8–12], shear force (e.g., silicones or

fluoropolymers with low surface energy) [13–15], lubricant infusion and so on [16–18]. Nonetheless, there are still problems and challenges to be solved such as poor mechanical stability, poor static antifouling performance, and short-term antifouling durability of the coatings [19–21]. It is of great significance to actively explore various eco-friendly and effective strategies to resist biofouling.

Reactive oxygen species (ROS), such as superoxide, hydrogen peroxide, and hydroxyl radicals, are highly reactive byproducts of oxygen formed during the process of aerobic metabolism in cells [22]. Although there is limited direct evidence, some reports indirectly suggest that ROS may be a known factor in the adhesion of many marine fouling organisms. For

example, as an important signal molecule of microorganisms, the excessive accumulation of ROS will lead to oxidative stress, which plays an important role in the production of extracellular polymeric substances (EPS) [23–25]. It is well known that EPS is a favorable guarantee for fouling organisms such as marine bacteria and diatoms to form biofilms [26]. In addition, the surface settlement of mussels is closely related to the 3,4-dihydroxy phenyl-L-alanine (DOPA) functional element, a key component in the structure of mussel adhesion proteins [27]. In the presence of oxygen, DOPA can spontaneously carry out oxidation, self-polymerization, and cross-linking reactions to generate polydopamine (PDA), which is crucial for the surface adhesion of mussels [28–31]. The previous reports mention that redox insulation and hypoxia weaken mussel attachment [32, 33]. Once fouling organisms form biofilms, they will increase the difficulty of antifouling work and bring more serious negative impacts [34–37]. However, in complex marine environments, the mechanism of fouling organism anti-adhesion and development of coating is still rare.

Antioxidants have excellent ROS scavenging characteristics and may become a potential new strategy for anti-biofilm, which is widely used in the biomedical field [23, 38, 39]. As a natural polyphenol antioxidant, tannic acid (TA) has a wide range of sources, biosafety, and exists in various plants [40, 41]. TA in nature can help plants resist the invasion of marine bacteria (such as TA in mangrove bark) [42], and commercial TA also has good antibacterial properties [41]. Previous reports have also shown that the TA structure contains numerous catechol, pyrogallol, and aromatic ring structures, which not only endow it with strong ROS scavenging ability [43, 44], but also enable TA to bind to various surfaces through various interactions, such as hydrogen bonds, ion bonds, π - π /cation- π interactions [40]. The above evidence suggests that using TA to remove ROS may affect the surface adhesion of fouling organisms. Therefore, based on the motivation of preventing fouling organisms from adhering without damaging them, this paper fully utilized the above characteristics of TA to co-deposit DOPA and polyethyleneimine (PEI) on the surface of polyvinyl chloride (PVC) to obtain a relatively stable transition layer, and then TA molecules were grafted to the surface to prepare a TA functionalized PVC (PVC/(PDA/PEI)-TA) coating. The effects of this coating on the surface adhesion of marine bacteria, benthic diatoms, and marine mussels were investigated. This study provides a new idea for the development of eco-friendly coatings for marine applications.

2 | Materials and Methods

2.1 | Chemicals

All chemical reagents are analytical grade and have not been further purified. Hydrochloric acid, dopamine hydrochloride, and ethylenediamine-terminated polyethyleneimine were purchased from Shanghai Aladdin Biochemical Technology Co. Ltd. Absolute ethanol (purity $\geq 99.7\%$) was purchased from Tianjin Fuyu Fine Chemical Co. Ltd. Tris(hydroxymethyl) aminomethane was purchased from Shanghai Macklin Biochemical Co. Ltd. Tannic acid was purchased from Tianjin Dengfeng Chemical Reagent Factory. Benthic diatoms were purchased from the Chinese Academy of Sciences Qingdao

Algae Germplasm Resources Bank. FePO₄ was purchased from Sinopharm Chemical Reagent Factory. Peptone, yeast extract, and agar were purchased from Beijing Boxing Biotechnology Co. Ltd.

2.2 | Preparation of Solution

Tris(hydroxymethyl) aminomethane was added to an appropriate amount of deionized water to prepare a buffer solution. After stirring for a while, it was titrated with a dilute hydrochloric acid solution with a concentration of 0.1 M to pH = 8.5. Then, an appropriate amount of dopamine hydrochloride was added to the buffer solution to prepare the DOPA (2.0 g/L) solution. Different concentrations of PEI were added to the prepared DOPA solution to prepare mixed DOPA and PEI solutions with different molar ratios. An appropriate amount of tannic acid was added to deionized water and thoroughly stirred to complete dissolution to obtain a 10 mM tannic acid solution.

2.3 | Preparation of PVC/(PDA/PEI)-TA Coatings

Several PVC boards purchased online with a thickness of 1 mm were cut into different sizes to match experimental requirements. Then, the PVC boards were gently wiped with alcohol cotton and placed in an ultrasonic cleaner for 10 min to remove residual impurities from the surfaces. The cleaned boards were placed in an oven and dried at a low temperature and then taken out for standby. Then, the dried PVC boards were immersed in an equal volume of a mixed solution of DOPA and PEI for 24 h, and the concentration ratio of DOPA to PEI is 10:1. Note that the PVC boards should be placed at the bottom of the weighing bottles so that they are completely immersed in the solution. Then, the coating surfaces were slowly and repeatedly cleaned with deionized water to remove molecules that were not firmly bonded to the surfaces. Finally, these coatings were immersed in a pre-prepared tannic acid solution (10 mM) for 24 h, and the surfaces were cleaned with deionized water. After drying with N₂, they were sealed and stored for standby.

2.4 | Characterization

The UV absorbance of the coating surface was measured by a TU-1901 UV spectrophotometer, and the scanning wavelength range was 800–200 nm. The water contact angle of the sample surfaces was tested by the JC2000D contact angle measuring instrument. The three-dimensional morphology of the coatings was observed by an OLS4000 laser confocal microscope. The functional group structure of coating surfaces was analyzed and detected using a Frontier Fourier transform infrared spectroscopy analyzer. The scanning range was 4000–650 cm⁻¹ and the resolution was 4 cm⁻¹.

2.5 | Stability Testing of Coatings

In this paper, two methods were used to test the acid–base stability of coatings. One method was to use absorbance. Because of its “melanin like” properties, PDA has good light

absorption. As PDA is easily dissociated under strong acid or alkali conditions immerse the coating in a strong acid/alkali solution and shake for 10 min. After cleaning, an appropriate amount of cleaning liquid was sucked up by the rubber tip droppers, and the absorbance of the cleaning liquid was tested. Another method was to use the changes in the wettability of the coating surface. The Water contact angle before and after immersing in a strong acid/alkali solution was measured and recorded, and the stability of the coating was characterized by the difference in the average contact angle of the coating before and after immersing.

2.6 | Study on Adhesion Behavior of Mussels

Mussels collected from the coast of the Yellow Sea were washed and placed in fresh seawater overnight to prepare for subsequent experiments. PVC/(PDA/PEI)-TA, glass slides (with high surface energy, high hardness, and smoothness), PVC, and polytetrafluoroethylene (PTFE, with extremely low surface energy and excellent antifouling performance) were laid on the bottom of the plastic tank and made into a 3 × 8 surface chessboard. Overall, 20 mussels were randomly placed on the chessboard. Then, the plastic tank was filled with fresh seawater. The location of the mussels was recorded after 48 h. When observing the dynamic and static adhesion behavior of mussels, four mussels were placed in glass culture dishes with PVC boards and PVC/(PDA/PEI)-TA coatings on the bottom. The HLD spiral test frame (HP-5, Digital push-pull force gauge, Yueqing Handpi Instruments Co. Ltd., China) was used to test the adhesion force between mussels and substrate surfaces.

2.7 | Assays of Marine Bacteria

All experimental equipment was sterilized by a pressure steam sterilization pot and then placed in a sterile box for standby. PVC boards and PVC/(PDA/PEI)-TA coatings were immersed in equal volumes of fresh seawater for 24 h, respectively, allowing bacteria to adhere and grow on the surfaces. Then, the samples were evenly divided into two groups, one group being rinsed in sterile seawater (50 mL) to remove bacteria that were not firmly attached to the surface. The other group was placed in a centrifuge tube filled with sterile seawater (50 mL) and oscillated at a frequency of 130 r/min for 20 min to simulate the scouring of the coating surface by water flow during ship navigation. After completion, the bacterial liquid was diluted to 10⁻⁶. The pipette gun was used to remove 10 μL bacterial liquid droplets and then uniformly disperse these bacterial liquid droplets in a 2216E solid culture medium. The colony status after 24 h of incubation in a biochemical incubator was recorded. Image-Pro Plus 6.0 software was used to calculate the adhesion rate of bacteria in a solid medium.

2.8 | Assays of Benthic Diatoms

The benthic diatom algal liquid was added to sterile aged seawater at a volume ratio of 1:2, and a nutrient salt prepared in advance was added thereto. The above solution was loaded into a weighing

bottle and cultured in a light incubator. After preliminary cultivation, the diatom species were transferred to PVC boards and PVC/(PDA/PEI)-TA coating surfaces and then placed in a light incubator for further cultivation. The coatings were evenly divided into two groups, and the coatings were rinsed and scoured separately after 48 h. Then, acetone (90%) was used to extract chlorophyll attached to the coating surfaces, which was generated by benthic diatoms as they grow on the coating surfaces. The extraction liquid was centrifuged using a centrifuge at a rotational speed of 4000 r/min. Then, the ultraviolet absorbance of the supernatant after centrifugation was tested. The concentration of chlorophyll was calculated according to the formula:

$$\rho_1 = 11.85(A_{664} - A_{750}) - 1.54(A_{647} - A_{750}) - 0.08(A_{630} - A_{750}).$$

where ρ_1 was the mass concentration of chlorophyll in the sample, A_{664} , A_{750} , A_{647} , and A_{630} were the absorbance values of the sample at wavelengths of 664, 750, 647, and 630 nm, respectively.

3 | Results and Discussion

3.1 | Surface Preparation

It is well-known that a tightly adhered PDA layer can be formed on nearly all materials (e.g., metals, ceramics, polymers) through a simple dip-coating procedure in dopamine aqueous solution [45]. This surface modification technique was facile and versatile. The resulting PDA coating was easy to further surface-tailor and functionalize through secondary treatments and had rapidly penetrated a wide range of research fields [46–48]. However, PDA molecules were mostly connected by non-covalent bonds, making them unstable and prone to dissociation and detachment in strong acid and alkali environments [45]. In this paper, PEI was introduced to overcome the disadvantage of poor stability of PDA coatings. As shown in Figure 1, by using the co-deposited DOPA and PEI, an intermediate transition layer (PVC/(PDA/PEI)) was preferentially created on the PVC surface. Then, a PVC/(PDA/PEI)-TA coating was prepared by grafting TA onto the surface of the coating through simple solution immersion.

3.2 | Effect of Molar Ratio of DOPA/PEI Co-Deposited on Coating Surfaces

The effects of DOPA/PEI co-deposited molar ratios on the coating surfaces were investigated by fixing the DOPA concentration (2.0 g/L) and changing the amount of PEI added. The co-deposited coating surfaces were observed, and the results were shown in Figure 2. The color, absorbance, surface wettability, and three-dimensional morphology of the coating surfaces were affected by the DOPA/PEI molar ratios. When the molar ratio of DOPA/PEI was 10:1, the coating surface had the darkest color and the highest absorbance (Figure 2a). In addition, compared to PVC board (75.9°), the surface hydrophilicity of coatings with a molar ratio of 1:0 and 10:1 was further improved, with water contact angles of 53.3° and 35.6°, respectively. However, it was not that the more PEI was added, the higher the hydrophilicity of the coating. When the DOPA/PEI molar ratio reached 1:10,

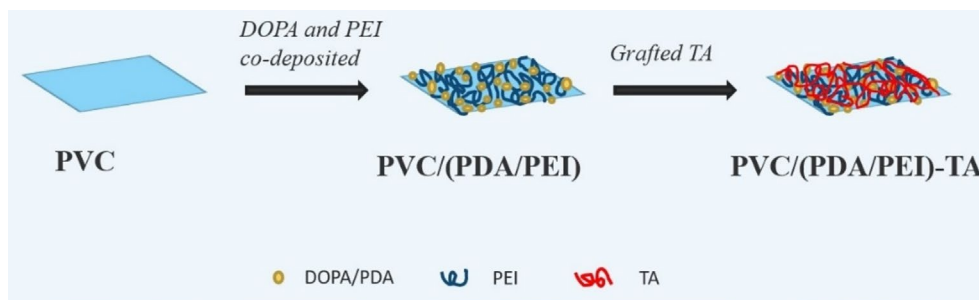


FIGURE 1 | Preparation process of PVC/(PDA/PEI)-TA coatings. PDA was mostly connected by noncovalent bonds between molecules, which were easy to dissociate and fall off in strong acid and alkali environments, resulting in poor stability of the coating. DOPA and PEI co-deposited can improve the stability of coatings. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

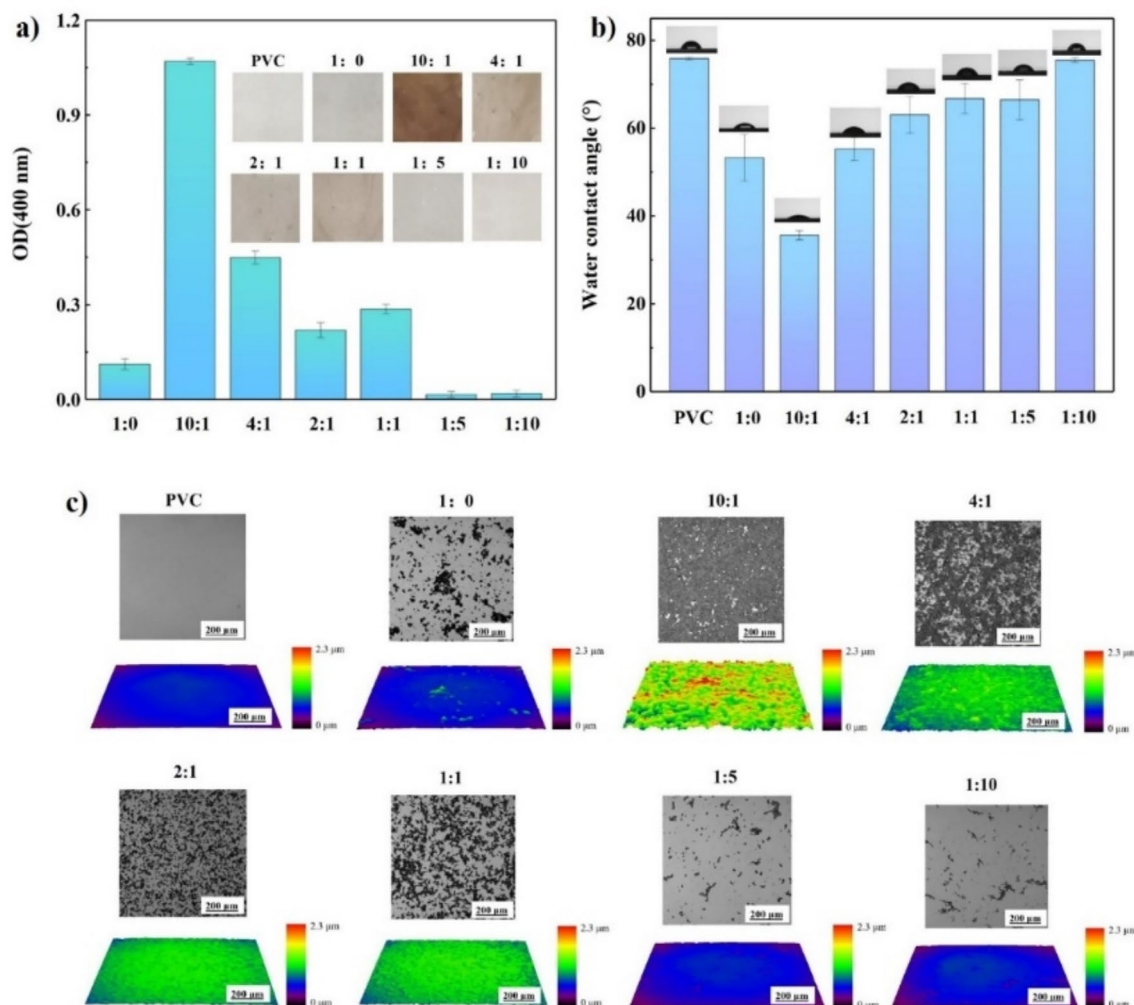


FIGURE 2 | Effect of DOPA/PEI molar ratios on the coatings. (a) Effect on the color and absorbance of the coating surfaces; (b) water contact angle showed the effect of DOPA/PEI molar ratios on the wettability of the coating surfaces; (c) laser confocal microscopy images showed the effect of DOPA/PEI molar ratios on the three-dimensional morphology of the coating surfaces. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

the water contact angle of the coating surface was similar to the PVC board (Figure 2b). As shown in Figure 2c, the three-dimensional morphology of the coating surfaces was observed.

The surface deposited with only PDA (the molar ratio was 1:0) produced only partial protrusions. In contrast, when the ratio of DOPA/PEI was 10:1, 4:1, 2:1, and 1:1, the uniformity of the coating

surfaces was significantly improved, and the co-deposited products were more evenly dispersed and covered the coating surfaces. When the molar ratio of DOPA/PEI was 10:1, the coating surface was covered with the thickest and most uniform deposits. However, with the further increase of PEI content (molar ratios were 1:5 and 1:10), almost no co-deposited products were formed on the coating surfaces.

The mechanism of the oxidative crosslinking reaction of DOPA had not yet been determined, and PEI made the reaction in the system more complex. The mechanism of the reaction between DOPA and PEI remained to be explored, but previous reports mentioned that a Michael addition reaction may occur between PEI and DOPA [49, 50]. From the above experimental results of this paper, it can be seen that a small amount of PEI addition may promote the surface deposition process of PDA. However, excessive PEI may disrupt non-covalent interactions in PDA aggregates (such as hydrogen bonds and π - π interaction) to inhibit the PDA deposition [51]. Therefore, it was very necessary to explore the optimal molar ratio of DOPA/PEI.

3.3 | Stability of the Coatings

To investigate the stability of the PDA/PEI co-deposited coating and select the optimal molar ratio of DOPA/PEI, the coatings were immersed in an equal volume of strong acid and strong alkali solution, respectively, and shaken for cleaning for 10 min. The absorbance of the solutions before and after cleaning and the water contact angle of the coating surfaces were tested. As shown in Figure 3, whether in strong acid or alkali solutions, the absorbance of DOPA/PEI co-deposited at a molar ratio of 10:1 and the water contact angle difference of the coating surfaces before and after cleaning were smaller than those with a molar ratio of 1:0 (only PDA deposition). Since PDA molecules were

mostly connected by non-covalent bonds, which were unstable and easily dissociated in strong acid and alkali environments, the detached PDA dissolved into the solution, resulting in an increase in the absorbance of the solution and a significant change in the water contact angle of the coating surface. At the molar ratio of 10:1, a small amount of PEI added enhanced the stability of the coating.

In summary, compared to other ratios, this paper selected the optimal molar ratio of DOPA/PEI co-deposited as 10:1 based on experimental needs. The co-deposited coating at this ratio had the following characteristics: more products were deposited on the surface, with high uniformity, the best hydrophilicity, and the most critical was the highest stability of the coating surface.

3.4 | Chemical Structure of the Coatings

As shown in Figure 4, the Fourier transform infrared spectrum of PVC showed three peaks in the range of 3100–2600 cm^{-1} . They appeared at 2959, 2913, and 2848 cm^{-1} and were attributed to $\nu(\text{C-H})$ vibrations [52]. After the deposition of PDA on the PVC surface, there was a difference from the PVC surface spectrum. There was only one peak in the range of 3100–2600 cm^{-1} , attributed to the asymmetrical stretching of H-Cl [53], and the asymmetric stretching of C-H of CH_2 at 2913 cm^{-1} disappeared [54]. However, its spectrum was generally similar to that of PVC,

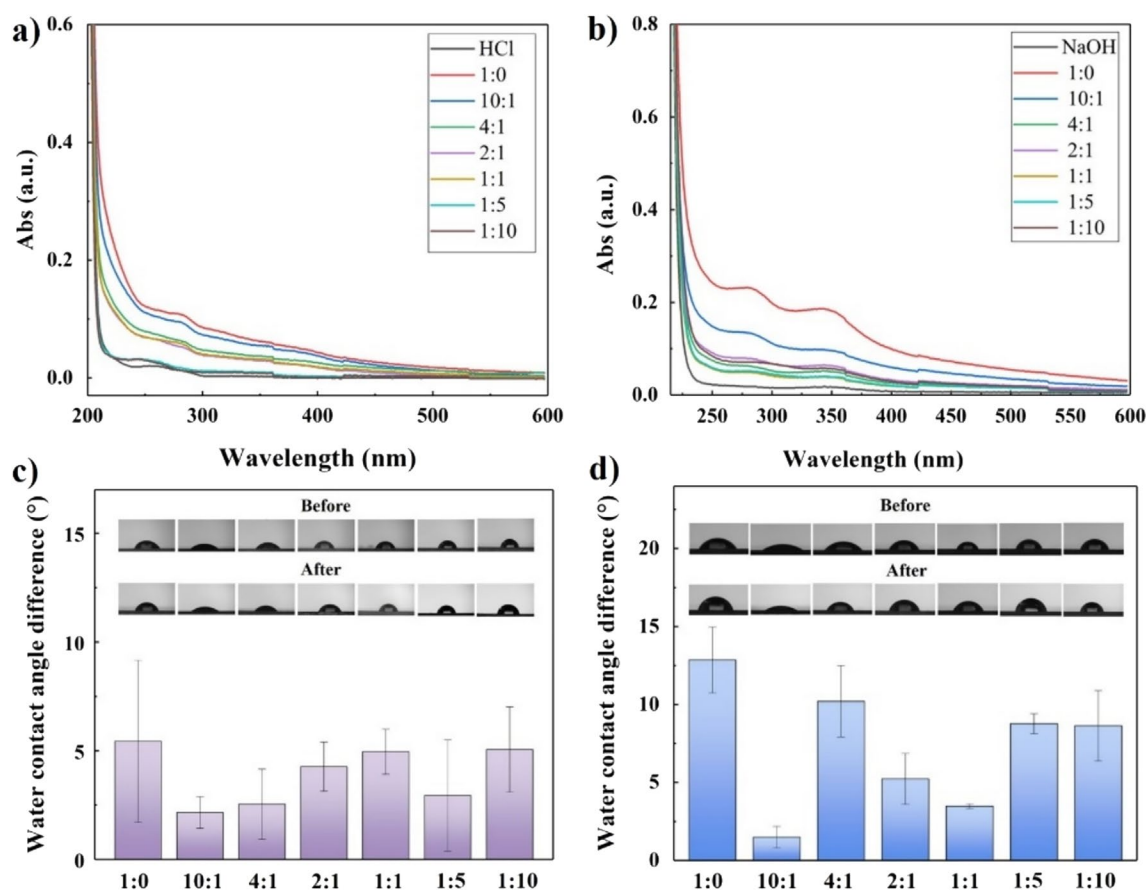


FIGURE 3 | Effect of DOPA/PEI molar ratios on the acid-base stability of coatings (a, b) were the absorbance of the solution after soaking the coating with HCl and NaOH for 10 min, respectively; (c, d) were the difference between the contact angles of the coating surfaces before and after immersion in HCl and NaOH, respectively. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.57227)]

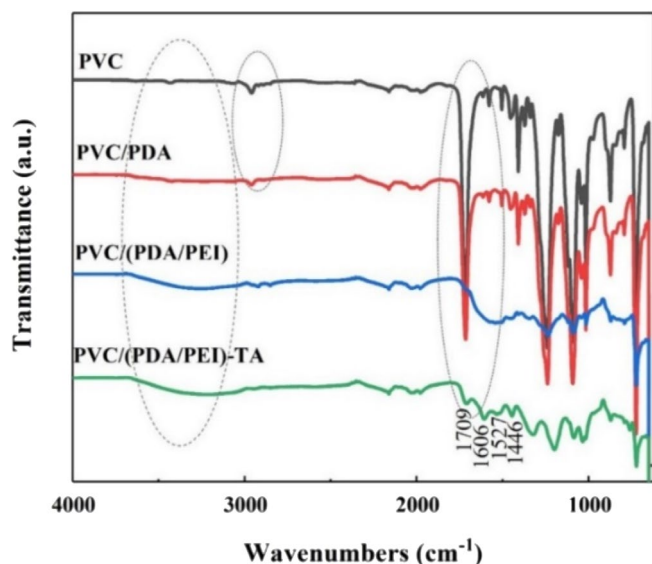


FIGURE 4 | Fourier infrared spectrum of coatings. The infrared spectrum of PVC/(PDA/PEI)-TA changed significantly, and the appearance of characteristic absorption peaks indicated that TA was successfully grafted onto the coating surfaces. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

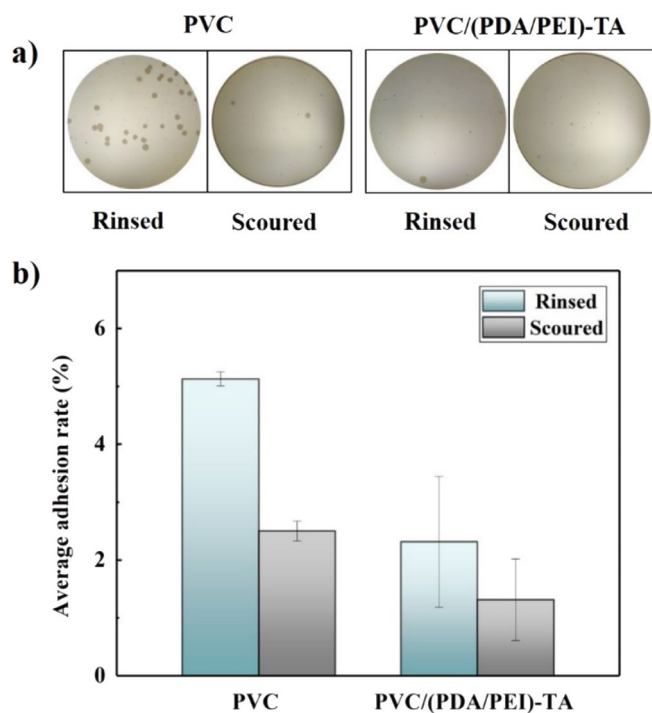


FIGURE 5 | The coatings were rinsed and scoured, respectively. Then the bacterial cells on the coating surfaces were transferred and cultured in a solid medium for 24 h. (a) Digital picture of marine bacterial adhesion; (b) average adhesion rate of marine bacteria. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

and we speculated that this may be due to the thinner PDA film deposited on the surface. The spectra of PVC/(PDA/PEI) and PVC/(PDA/PEI)-TA showed significant changes compared to those of PVC and PVC-PDA, with some absorption peaks significantly widened and a wider absorption peak (attributed to

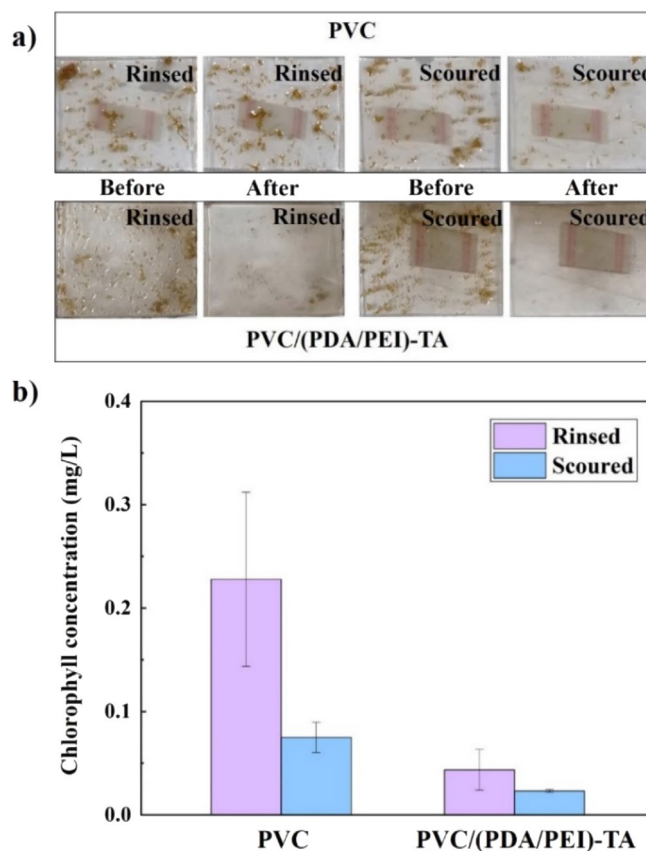


FIGURE 6 | The coatings were rinsed and scoured respectively after culturing benthic diatoms on the coating surfaces for 48 h. (a) Digital photos of benthic diatoms adhering to the coating surfaces before and after being rinsed and scoured; (b) chlorophyll concentration on the coating surfaces after being rinsed and scoured. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

the phenolic hydroxyl stretching mode) generated in the range of 3600–3100 cm^{-1} . Moreover, some absorption peaks within the range of 1800–1600 cm^{-1} disappeared, such as the C=C conjugated double bonds and carbonyl groups at 1720 and 1620 cm^{-1} [55–57]. The thick DOPA/PEI co-deposited products had formed on the PVC surface. Compared with the PVC/(PDA/PEI) surface spectrum, the PVC/(PDA/PEI)-TA spectrum had an obvious characteristic peak at 1709 cm^{-1} attributed to the stretching vibrations of C=O (carboxylic ester) groups, which were the characteristic functional group in the molecular structures of TA. In addition, peaks at 1606, 1527, and 1446 cm^{-1} were consistent with the aromatic C—C stretches [58]. It can be seen that TA had been successfully connected onto the surface.

3.5 | Effect of PVC/(PDA/PEI)-TA Surface on Adhesion of Marine Bacteria

Most marine bacteria are Gram negative. The common marine bacteria are mainly *Pseudomonas*, *Vibrio*, and so on [59]. To investigate the impact of TA on the adhesion of marine bacteria and preliminarily evaluate the antifouling performance of PVC/(PDA/PEI)-TA coating, PVC boards and PVC/(PDA/PEI)-TA coatings were rinsed and scoured, respectively, to simulate the erosion of water on the hull surface when the ship is berthing and sailing.

Then the bacterial solutions were diluted and transferred to 2216E solid medium and incubated in a biochemical incubator at 25°C for 24 h. The bacterial colony status after 24 h of cultivation was recorded. As shown in Figure 5a, the rinsed PVC surfaces had the most bacterial colonies, while the PVC/(PDA/PEI)-TA surfaces had fewer bacterial colonies. The bacterial colony areas were quantified, and the results were shown in Figure 5b. The adhesion rate of marine bacteria on PVC/(PDA/PEI)-TA surfaces was lower than that on PVC surfaces, regardless of whether they were rinsed or scoured. These results indicated that surface grafting TA can significantly reduce the adhesion rate of marine bacteria.

3.6 | Effect of TA on the Adhesion of Benthic Diatoms

PVC boards and PVC/(PDA/PEI)-TA coatings with the same specification were immersed in an equal volume of algae solution and then incubated in a light incubator for 24 h. After 24 h, the coatings were rinsed and scoured respectively, and photos were taken to record the adhesion status of benthic diatoms. As shown in Figure 6a, there were many benthic diatoms attached to the PVC surfaces, and after being rinsed and scoured, there were still many benthic diatoms remaining on the surfaces. However, after rinsing and scouring the PVC/(PDA/PEI)-TA coatings, almost all benthic diatoms on the surfaces fell off. The remaining chlorophyll on the coating surfaces was extracted, and the concentration of chlorophyll was calculated

to characterize the coating's resistance to benthic diatom adhesion. The lower the chlorophyll concentration, the easier it was for benthic diatoms to fall off the surfaces, and the better the antifouling performance of the coatings. As shown in Figure 6b, the chlorophyll concentration on the PVC/(PDA/PEI)-TA coating surfaces was much lower than that on the PVC surfaces. It can be seen that PVC/(PDA/PEI)-TA coatings had excellent resistance to the adhesion of benthic diatoms.

3.7 | Effect of TA on Adhesion Behavior of Mussels

To explore the effect of TA on the adhesion behavior of mussels, PVC/(PDA/PEI)-TA, glass slides, PVC, and PTFE substrates were laid at the bottom of the plastic tank in the order shown in Figure 7a. These substrates were marked from 1 to 24 to make a 3×8 surface chessboard. Then, 20 mussels were randomly placed on a chessboard and injected with fresh seawater in the tank (Figure 7b). After 48 h, the locations of the mussels were photographed and recorded to evaluate their preference for the substrates. Figure 7c reported the selective adhesion of mussels to substrates. Only a small number of mussels chose to adhere to the TA coating surfaces. The adhesive plaques on the several coating surfaces (Marked 10–14 and 18–22) with the highest concentration of mussels were circled with blue outlines. As shown in Figure 7d, it was observed that mussels secreted the most adhesive plaques on glass slides (Marked 12 and 21) and PVC surfaces (No. 19), and a small number of plaques were also observed

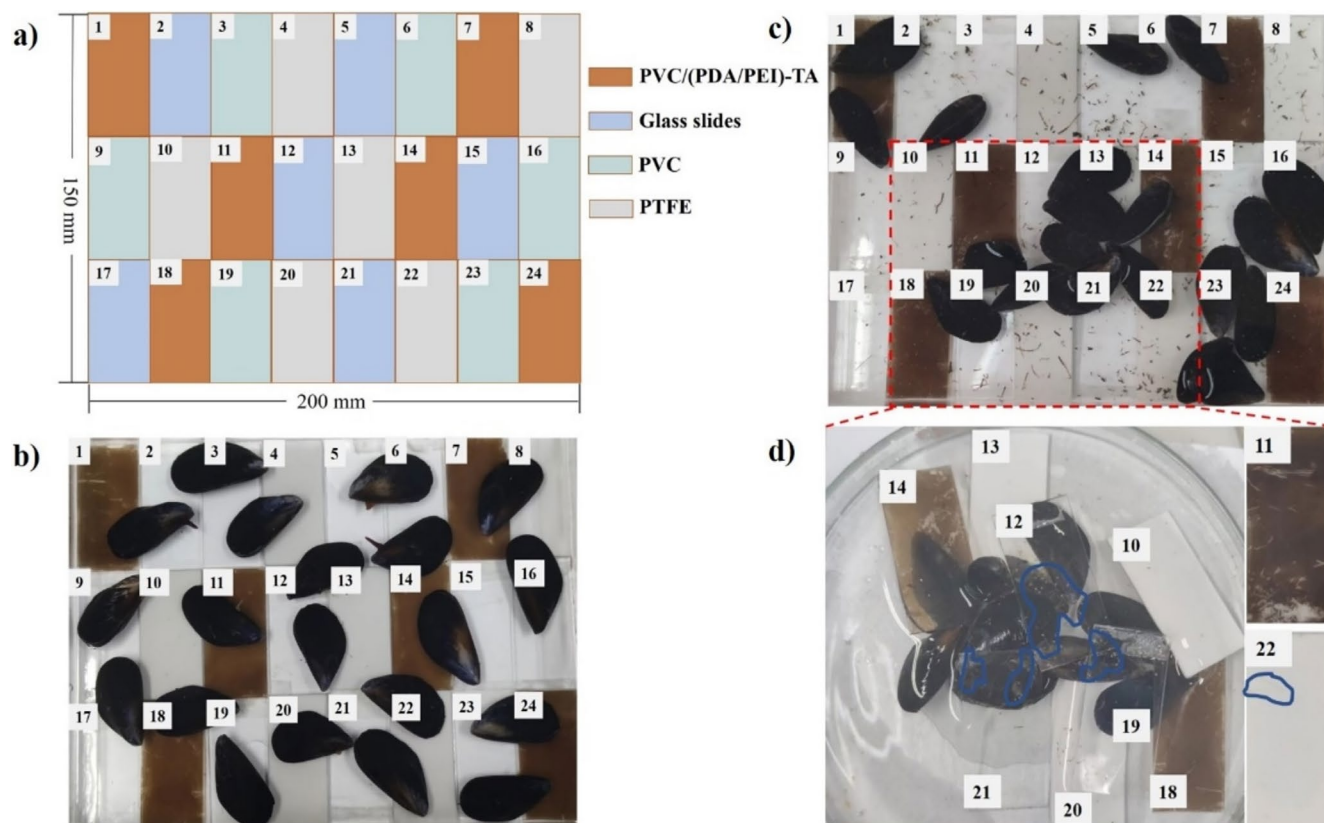


FIGURE 7 | The surface checkerboard was prepared with various coatings to investigate the selective adhesion behavior of mussels to the substrate. (a) Schematic diagram of a surface checkerboard; (b) twenty mussels were randomly placed on the chessboard; (c) digital photos of mussels' behavior on the chessboard after 48 h; (d) adhesion plaques on the coatings where mussels gather most. (The blue outlines highlight plaques). [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.57227)]

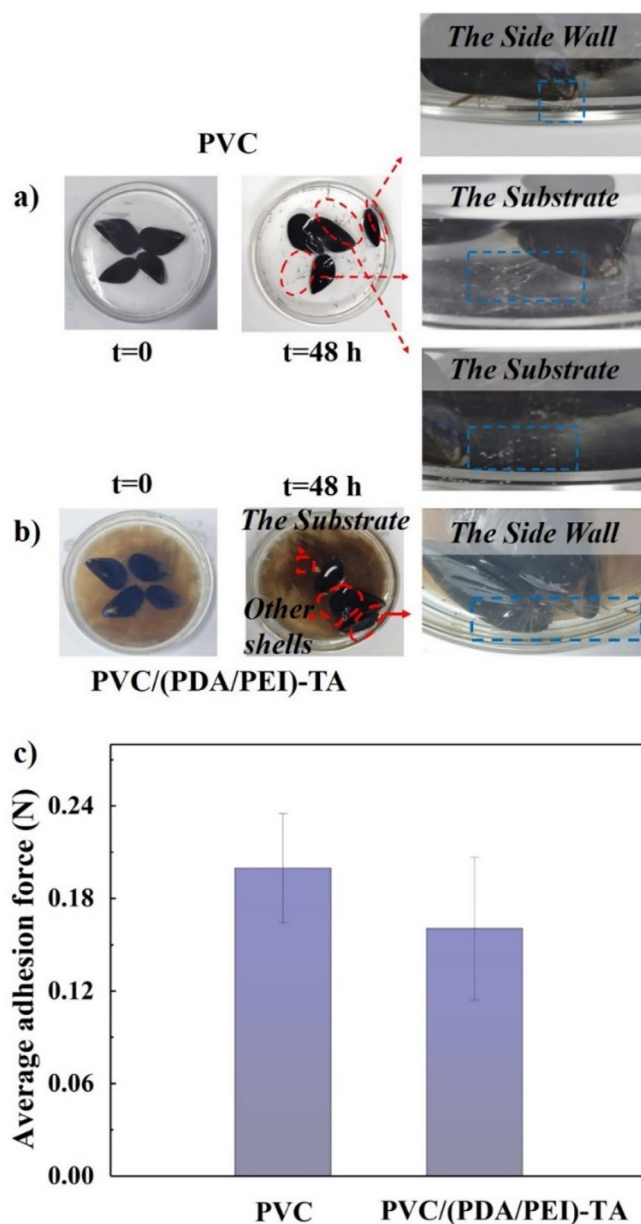


FIGURE 8 | Adhesion behavior of mussels on (a) PVC coated surface; (b) PVC/(PDA/PEI)-TA coated surface; (c) average adhesion force between plaques and coatings. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

on PTFE (No. 22) and TA coating surfaces (No. 14). The lower surface energy of PTFE may be the reason for the lower adhesion rate of mussels. On the surface of PVC/(PDA/PEI)-TA, there were also fewer adhesive plaques of mussels, which we believed may be due to TA interference with mussels' substrate selection. Therefore, PVC board and PVC/(PDA/PEI)-TA coating were selected to further investigate the effect of TA on the surface adhesion of mussels.

The mussels were placed in the glass culture dishes with PVC and PVC/(PDA/PEI)-TA coatings at the bottom. Then, the adhesion behaviors of the mussels were monitored and observed in real time. As shown in Figure S1, once the mussel explored the PVC board, it continuously secreted multiple byssal threads on the surface. However, the mussel on the surface of PVC/(PDA/PEI)-TA

quickly lifted its foot after lightly touching the surface. And after repeating the above process multiple times, the mussel retracted its foot into its shell and did not secrete byssal threads on the surface for a short time. The adhesion behaviors of mussels on PVC and PVC/(PDA/PEI)-TA surfaces were recorded after 48 h. As shown in Figure 8a, some mussels chose to adhere to similar shells, only a small number of plaques appeared on the side wall of the culture dish, leaving a large number of byssal threads and adhesive plaques on the PVC surface. There were significant differences in the PVC/(PDA/PEI)-TA surface. As shown in Figure 8b, adhesive plaques were more present on the shells of mussels and sidewall, while adhesive plaques were less present on the substrate. In addition, as shown in Figure 8c, the average adhesion force of mussels on the PVC/(PDA/PEI)-TA surface was smaller than that on the PVC surface. The above results indicated that TA affected the surface adhesion behavior of mussels and reduced the surface adhesion force. The mussels tended to secrete byssal threads on other surfaces rather than TA-coated surfaces.

4 | Conclusions

We demonstrated that the PVC/(PDA/PEI)-TA coatings had excellent antifouling performance. Not only could they resist the attachment of marine bacteria and benthic diatoms, but these fouling organisms were also prone to detachment from the surfaces under the erosion of water flow. Compared with PVC boards, the adhesion behavior of mussels on the PVC/(PDA/PEI)-TA coating surfaces has undergone significant changes. The mussel quickly lifted after lightly touching the surface with its foot, without continuously secreting threads after first exploring the surface. In addition, mussels were not easily able to adhere to the TA surface, and even if they adhered to the surface, the adhesion force between their plaques and the substrate was weak. The antifouling strategy based on tannic acid in this paper has the advantages of low toxicity, eco-friendliness, and freedom from substrate limitations, which may play a positive role in the design and development of antifouling coatings.

Author Contributions

Meihua Jin: methodology (lead). **Jine Wang:** writing – original draft (lead). **Rui Zhao:** data curation (lead). **Zhe Cheng:** data curation (equal). **Pei Wang:** formal analysis (lead). **Jiujun Xu:** methodology (equal).

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Ethics Statement

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that supports the findings of this study are available in the supplementary material of this article.

References

1. S. Pouhashem, A. Seif, F. Saba, et al., "Antifouling Nanocomposite Polymer Coatings for Marine Applications: A Review on Experiments, Mechanisms, and Theoretical Studies," *Journal of Materials Science and Technology* 118 (2022): 73–113.
2. G. D. Bixler and B. Bhushan, "Biofouling: Lessons From Nature, Philos," *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* 370 (2012): 2381–2417.
3. S. C. Brand, A. G. Jeffs, U. von Ammon, and A. Zaiko, "Assessing the Presence, Settlement and Growth of the Invasive Mediterranean Fanworm, *Sabella spallanzanii*, on Mussel Farms," *Journal of Experimental Marine Biology and Ecology* 554 (2022): 151767.
4. M. S. Selim, M. A. Shenashen, S. A. El-Safty, et al., "Recent Progress in Marine Foul-Release Polymeric Nanocomposite Coatings," *Progress in Master's Science* 87 (2017): 1–32, <https://doi.org/10.1016/j.pmatsci.2017.02.001>.
5. S. Hao, Y. H. QI, and Z. P. Zhang, "Settlement Behavior and Mechanism of *Navicula* sp. on WLAP/PDMS Composite Coating Under Simulative Diurnal Alternation, Constant Light and Dark Conditions," *Surface Science* 42 (2023): 103527, <https://doi.org/10.1016/j.surfsc.2023.103527>.
6. T. Mao, G. Lu, C. Xu, H. Yu, and J. Yu, "Preparation and Properties of Polyvinylpyrrolidone-Cuprous Oxide Microcapsule Antifouling Coating," *Progress in Organic Coating* 141 (2020): 105317.
7. R. Xie, X. Ai, Q. Xie, C. Ma, and G. Zhang, "Non-Silicone Elastic Coating With Fouling Resistance and Fouling Release Abilities Based on Degradable Hyperbranched Polymer," *Progress in Organic Coating* 175 (2023): 107350.
8. C. A. Del Grosso, C. Leng, K. Zhang, et al., "Surface Hydration for Antifouling and Bio-Adhesion," *Chemical Science* 11 (2020): 10367–10377.
9. Z. Liu, Z. Qiu, and C. He, "A Durable *Watasenia scintillans*-Like Luminescent Hydrophilic Coating for Synergistic High-Efficiency Anti-Biofouling," *Applied Surface Science* 588 (2022): 152968.
10. I. Banerjee, R. C. Pangule, and R. S. Kane, "Antifouling Coatings: Recent Developments in the Design of Surfaces That Prevent Fouling by Proteins, Bacteria, and Marine Organisms," *Advanced Materials* 23 (2011): 690–718.
11. J. Yang, H. Qiang, J. Wang, et al., "Mechanically Durable Antibacterial Nanocoatings Based on Zwitterionic Copolymers Containing Dopamine Segments," *Journal of Materials Science and Technology* 89 (2020): 233–241.
12. Y. Wei, X. Cao, J. Hu, et al., "Preparation of Salt-Tolerant Chitosan Hydrogels and Their Anti-Biofouling Behavior Study," *Surface and Coatings Technology* 460 (2023): 129403, <https://doi.org/10.1016/j.surfcoat.2023.129403>.
13. P. Shivapooja, Q. Yu, B. Orihuela, et al., "Modification of Silicone Elastomer Surfaces With Zwitterionic Polymers: Short-Term Fouling Resistance and Triggered Biofouling Release," *ACS Applied Materials & Interfaces* 7 (2015): 25586–25591.
14. P. Hu, X. Xie, C. Ma, and G. Zhang, "Silicone-Based Fouling-Release Coatings for Marine Antifouling," *Langmuir* 36 (2020): 2170–2183.
15. J. Xu, Y. Qi, and Z. Zhang, "Effects of 3,3,3-Trifluoropropyl Content in Poly (Dimethyl-Methylphenyl-Methyltrifluoropropyl) Siloxane Coating on Its Antifouling Performance," *Progress in Organic Coating* 197 (2024): 108777.
16. S. Amini, S. Kolle, L. Petrone, et al., "Preventing Mussel Adhesion Using Lubricant-Infused Materials," *Science* 357 (2017): 668–673.
17. H. Zhai, X. Li, Y. Liu, et al., "Di-Particles-Derived Slippery Lubricant-Infused Porous Surface With Broad Anti-Adhesion Performance," *Applied Surface Science* 616 (2023): 156462.
18. X. Xu, R. Chen, Y. Zheng, et al., "Slippery-Liquid-Infused Electrostatic Flocking Surfaces for Marine Antifouling Application," *Langmuir* 37 (2021): 10020–10028.
19. A. K. Leonardi and C. K. Ober, "Polymer-Based Marine Antifouling and Fouling Release Surfaces: Strategies for Synthesis and Modification," *Annual Review of Chemical and Biomolecular Engineering* 10 (2019): 241–264.
20. A. M. C. Maan, A. H. Hofman, W. M. Vos, and M. Kamperman, "Recent Developments and Practical Feasibility of Polymer-Based Antifouling Coatings," *Advanced Functional Materials* 30, no. 32 (2020): 2000936, <https://doi.org/10.1002/adfm.202000936>.
21. J. Chen, R. Jian, K. Yang, et al., "Urushiol-Based Benzoxazine Copolymer With Low Surface Energy, Strong Substrate Adhesion and Antibacterial for Marine Antifouling Application," *Journal of Cleaner Production* 318 (2021): 128527.
22. P.-L. Lam, R. S. M. Wong, K.-H. Lam, et al., "The Role of Reactive Oxygen Species in the Biological Activity of Antimicrobial Agents: An Updated Mini Review," *Chemical-Biological Interactions* 320 (2020): 109023, <https://doi.org/10.1016/j.cbi.2020.109023>.
23. K. S. Ong, C. I. Mawang, D. Daniel-Jambun, Y. Y. Lim, and S. M. Lee, "Current Anti-Biofilm Strategies and Potential of Antioxidants in Biofilm Control," *Expert Review of Anti-Infective Therapy* 16 (2018): 855–864.
24. M. Gambino and F. Cappitelli, "Mini-Review: Biofilm Responses to Oxidative Stress," *Biofouling* 32 (2016): 167–178.
25. J. Wang, C. Zhang, R. Zhao, P. Wang, M. Jin, and J. Xu, "Antioxidant N-Acetylcysteine Removing ROS: An Antifouling Strategy Inspired by Mussels," *Environmental Science: Processes & Impacts* 25, no. 12 (2023): 1962–1973, <https://doi.org/10.1039/d3em00191a>.
26. H. Jin, L. Tian, W. Bing, J. Zhao, and L. Ren, "Bioinspired Marine Antifouling Coatings: Status, Prospects, and Future," *Progress in Materials Science* 124 (2022): 100889.
27. E. Astoricchio, C. Alfano, L. Rajendran, P. A. Temussi, and A. Pastore, "The Wide World of Coacervates: From the Sea to Neurodegeneration," *Trends in Biochemical Sciences* 45 (2020): 706–717.
28. J. Saiz-Poseu, J. Mancebo-Aracil, F. Nador, F. Busqué, and D. Ruiz-Molina, "The Chemistry Behind Catechol-Based Adhesion," *Angewandte Chemie, International Edition* 58 (2019): 696–714.
29. C. Zhang, L. Xiang, J. Zhang, et al., "Revisiting the Adhesion Mechanism of Mussel-Inspired Chemistry," *Chemical Science* 13, no. 6 (2022): 1698–1705, <https://doi.org/10.1039/d1sc05512g>.
30. D. R. Miller, J. E. Spahn, and J. H. Waite, "The Staying Power of Adhesion-Associated Antioxidant Activity in *Mytilus californianus*," *Journal of the Royal Society Interface* 12 (2015): 20150614.
31. J. Yang, M. A. C. Stuart, and M. Kamperman, "Jack of all Trades: Versatile Catechol Crosslinking Mechanisms," *Chemical Society Reviews* 43 (2014): 8271–8298.
32. M. N. George, B. Pedigo, and E. Carrington, "Hypoxia Weakens Mussel Attachment by Interrupting DOPA Cross-Linking During Adhesive Plaque Curing," *Journal of the Royal Society Interface* 15 (2018): 20180489.
33. E. Valois, R. Mirshafian, and J. H. Waite, "Phase-Dependent Redox Insulation in Mussel Adhesion," *Science Advances* 6 (2020): eaaz6486.
34. N. Tang, S. Yuan, Y. Luo, et al., "Nanoparticle-Based Photodynamic Inhibition of *Candida albicans* Biofilms With Interfering Quorum Sensing," *ACS Omega* 8, no. 4 (2023): 4357–4368, <https://doi.org/10.1021/acsomega.2c07740>.
35. A. Omar, J. B. Wright, G. Schultz, R. Burrell, and P. Nadworny, "Microbial Biofilms and Chronic Wounds," *Microorganisms* 5, no. 1 (2017): 9.

36. E.-H. Lee, S.-W. Lee, Y. Seo, et al., "Manganese Oxide Nanozyme-Doped Diatom for Safe and Efficient Treatment of Peri-Implantitis," *ACS Applied Materials & Interfaces* 14 (2022): 27634–27650.
37. Y. Oz, A. Nabawy, S. Fedeli, et al., "Biodegradable Poly(Lactic Acid) Stabilized Nanoemulsions for the Treatment of Multidrug-Resistant Bacterial Biofilms," *ACS Applied Materials & Interfaces* 13 (2021): 40325–40331.
38. A. Bouyahya, D. Taha, T. Benali, et al., "Natural Sources, Biological Effects, and Pharmacological Properties of Cymaroside," *Biomedicine & Pharmacotherapy* 161 (2023): 114337.
39. Y. Zhang, Y. Qian, M. Zhang, and W. Qiao, "Repairing of Rutin to the Toxicity of Combined F-53B and Chromium Pollution on the Biofilm Formed by *Pseudomonas aeruginosa*," *Journal of Environmental Sciences* 127 (2023): 158–168.
40. C. Chen, H. Yang, X. Yang, and Q. Ma, "Tannic Acid: A Crosslinker Leading to Versatile Functional Polymeric Networks: A Review," *RSC Advances* 12 (2022): 7689–7711.
41. H. Xu, W. Huang, K. Ren, and Y. Tang, "Spraying Layer-By-Layer Assembly of Tannin-Fe³⁺ and Polyethyleneimine for Antibacterial Coating," *Colloids and Surfaces* 42 (2021): 100422, <https://doi.org/10.1016/j.colcom.2021.100422>.
42. M. Cui, Z. Wang, and B. Wang, "Survival Strategies of Mangrove (*Ceriops tagal* (Perr.) C. B. Rob) and the Inspired Corrosion Inhibitor," *Frontiers in Materials* 9 (2022): 879525, <https://doi.org/10.3389/fmats.2022.879525>.
43. Z. Ni, H. Yu, L. Wang, et al., "Multistage ROS-Responsive and Natural Polyphenol-Driven Prodrug Hydrogels for Diabetic Wound Healing," *ACS Applied Materials & Interfaces* 14 (2022): 52643–52658.
44. N. P. Bona, M. S. P. Soares, N. S. Pedra, et al., "Tannic Acid Attenuates Peripheral and Brain Changes in a Preclinical Rat Model of Glioblastoma by Modulating Oxidative Stress and Purinergic Signalling," *Neurochemical Research* 47 (2022): 1541–1552.
45. W. Yang, C. Liu, and Y. Chen, "Stability of Polydopamine Coatings on Gold Substrates Inspected by Surface Plasmon Resonance Imaging," *Langmuir* 34 (2018): 3565–3571.
46. J. Huang, W. Zhang, H. Li, X. Yu, S. Ding, and C. Wu, "An Autonomous Self-Healing Hydrogel With High Polydopamine Content for Improved Tensile Strength," *Journal of Materials Science* 55 (2020): 17255–17265.
47. T. M. D'Alvise, S. Harvey, L. Hueske, et al., "Ultrathin Polydopamine Films With Phospholipid Nanodiscs Containing a Glycophorin a Domain," *Advanced Functional Materials* 30, no. 21 (2020): 2000378, <https://doi.org/10.1002/adfm.202000378>.
48. Y. Zheng, X. Wang, and G. Wu, "Facile Strategy of Improving Interfacial Strength of Silicone Resin Composites Through Self-Polymerized Polydopamine Followed via the Sol-Gel Growing of Silica Nanoparticles Onto Carbon Fiber," *Polymers* 11 (2019): 1639.
49. S. Hong, C. F. Schaber, K. Dening, E. Appel, S. N. Gorb, and H. Lee, "Air/Water Interfacial Formation of Freestanding, Stimuli-Responsive, Self-Healing Catecholamine Janus-Faced Microfilms," *Advanced Materials* 26 (2014): 7581–7587.
50. C. Zhao, F. Zuo, Z. Liao, Z. Qin, S. Du, and Z. Zhao, "Mussel-Inspired One-Pot Synthesis of a Fluorescent and Water-Soluble Polydopamine-Polyethyleneimine Copolymer," *Macromolecular Rapid Communications* 36, no. 10 (2015): 909–915, <https://doi.org/10.1002/marc.201500021>.
51. D. R. Dreyer, D. J. Miller, B. D. Freeman, D. R. Paul, and C. W. Bielawski, "Elucidating the Structure of Poly(Dopamine)," *Langmuir* 28 (2012): 6428–6435.
52. S. Yang, C. Tran, G. S. Whiteley, et al., "Covalent Immobilization of N-Acetylcysteine on a Polyvinyl Chloride Substrate Prevents Bacterial Adhesion and Biofilm Formation," *Langmuir* 36 (2021): 13023–13033.
53. J. Zhou, G. Liu, S. Wang, H. Zhang, and F. Xu, "TG-FTIR and Py-GC/MS Study of the Pyrolysis Mechanism and Composition of Volatiles From Flash Pyrolysis of PVC," *Journal of the Energy Institute* 93 (2020): 2362–2370.
54. P. A. B. Ranjana and B. Sundaresan, "A Study on the Complex Formation of Poly(Methyl Methacrylate), Poly(Vinyl Chloride) and Their Blend With Lithium Perchlorate Using FTIR Spectroscopy," *Macromolecular Symposia* 392, no. 1 (2020): 2000172, <https://doi.org/10.1002/masy.202000172>.
55. R. A. Zangmeister, T. A. Morris, and M. J. Tarlov, "Characterization of Polydopamine Thin Films Deposited at Short Times by Autoxidation of Dopamine," *Langmuir* 29 (2013): 8619–8628.
56. X. Zhao, Y. Zhang, H. Hu, et al., "Effect of Lignin Esters on Improving the Thermal Properties of Poly(Vinyl Chloride)," *Journal of Applied Polymer Science* 136 (2019): 47176.
57. R. Sagar, M. S. Gaur, and A. A. Rogachev, "Piezoelectric and Pyroelectric Properties of Ceramic Nanoparticles Based Nanostructured PVDF/PVC Blend Nanocomposites," *Journal of Thermal Analysis and Calorimetry* 146, no. 2 (2020): 645–655, <https://doi.org/10.1007/s10973-020-09979-z>.
58. N. Ninan, A. Forget, V. P. Shastri, N. H. Voelcker, and A. Blencowe, "Antibacterial and Anti-Inflammatory pH-Responsive Tannic Acid-Carboxylated Agarose Composite Hydrogels for Wound Healing," *ACS Applied Materials & Interfaces* 8 (2016): 28511–28521.
59. C. Zhang, Y. Qi, Y. Guo, et al., "Anti-Marine Biofouling Adhesion Performance and Mechanism of PDMS Fouling-Release Coating Containing PS-PEG Hydrogel," *Marine Pollution Bulletin* 194 (2023): 115345.

Supporting Information

Additional supporting information can be found online in the Supporting Information section.